

MoES-NCMRWF / MOSDAC PS 26077 Data Foundation | Spiking Neural Net (SNN) Edge Gate |
2D Shallow Water Equations Inundation Coupling

2. Convective Cell Developing

4. PEAK CLOUDBURST & PINN TRIGGER (>100mm/hr)

⚡ CLOUDBURST LIKELY (EMERGENCY / PINN TRIGGERED)

71.4%

METEOROLOGICAL PARAMETER INFLUENCE BREAKDOWN (Normalized Contribution %)

- | | | | |
|------------------------------|------------------------|-------|----------------------------------|
| • Rain Intensity (R) | <div><div></div></div> | 45.2% | (val=124.8 mm/hr) |
| • Rain Acceleration (RI) | <div><div></div></div> | 26.0% | (val=+218.6 mm/hr ²) |
| • 30-min Accumulation (R30) | <div><div></div></div> | 11.8% | (val= 49.4 mm) |
| • 60-min Volume (R60) | <div><div></div></div> | 7.6% | (val= 50.7 mm) |
| • Humidity Surge (dRH) | <div><div></div></div> | 3.4% | (RH=98% (ΔRH=+10.0%)) |
| • GNSS Moisture Influx (IWV) | <div><div></div></div> | 7.5% | (ΔIWV=+4.8 mm) |

Station	Rain	Acc	SNN	State	Interval	Alert Tier
Name	Rate	(RI)	Membrane			

Khanpara (RSC)	2.7 mm/h	+10.4 mm/h ²	0.196/1.0	DORMANT	15m	DEVELOPING
North Guwahati	1.6 mm/h	-5.5 mm/h ²	0.143/1.0	DORMANT	15m	DEVELOPING

Station Name	Rain Rate	Acc (RI)	SNN Membrane	State	Interval	Alert Tier
Rangiya	4.4 mm/h	+7.0 mm/h ²	0.165/1.0	DORMANT	15m	DEVELOPING
Nalbari	101.6 mm/h	+177.6 mm/h ²	⚡ V=0.907/1.0	ACTIVE	5m	CLOUDBURST_LIKELY
Gossaigaon	0.8 mm/h	-15.5 mm/h ²	0.347/1.0	DORMANT	15m	DEVELOPING
Arunachal	124.8 mm/h	+218.6 mm/h ²	⚡ SPIKE FIRED!	ACTIVE	5m	CLOUDBURST_LIKELY



PINN HYDRODYNAMIC FLOOD INUNDATION SOLVER TRIGGERED!

CONVECTIVE EPICENTER

Lat 24.854°N, Lon 92.739°E
(Arunachal) & Lat 26.436°N, Lon 91.335°E (Nalbari)

PEAK RAIN CORE

124.8 mm/hr (Surpasses IMD
Cloudburst standard of >= 100 mm/hr)

SWE RUNOFF INFLUX (Q)

1,798,316.8 m³/s arriving
into catchment grid

SATELLITE ALIGNMENT

MOSDAC Kalpana-1 / INSAT-3D 4km
convective cooling tile synchronised

2D Shallow Water Equations Source Coupling:

$\partial h/\partial t + \partial(uh)/\partial x + \partial(vh)/\partial y = R(x,y,t) - I(x,y,t)$ | Catchment Mesh: 4km grid
spacing | Runoff excess feeding PINN forward solver.

Compiled from live simulation run | Antigravity AI Cloudburst Nowcasting Pipeline | Ready to share with team & evaluators